# Davide Boscaini, Ph.D.

*Teaching machines how to perceive and understand the surrounding environment*

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## About

I am a Research Scientist at the Technologies of Vision research unit of the Fondazione Bruno Kessler in Trento, Italy. My interests are in 3D perception and understanding, with a focus on object 6D pose estimation and 3D scene segmentation. According to Google Scholar, I have an h-index of 17, an i10-index of 24, and my articles have over 6,700 citations.

Before joining FBK, I received a PhD in Computational Science from the Università della Svizzera italiana in Lugano, Switzerland, in 2017. During my PhD, under the supervision of prof. Michael Bronstein, my research focused on extending deep learning techniques to geometric domains such as 3D shapes and graphs, contributing to the birth of a new research direction called Geometric Deep Learning. Prior to that, I obtained an M.S. in Mathematics from the University of Verona, Italy, in 2013, and a B.S. in Applied Mathematics from the same institution in 2010.

## Education

Ph.D. in Computational Science — Sep. 2013 – Sep. 2017
**Università della Svizzera italiana** — *Lugano, Switzerland*
Dissertation on "Geometric Deep Learning for Shape Analysis"
Advisor: M.M. Bronstein. Co-advisor: J. Masci
Examiners: J. Schmidhuber, M. Ovsjanikov, P. Vandergheynst, K. Hormann

M.S. in Mathematics — Oct. 2010 – Mar. 2013
**University of Verona** — *Verona, Italy*
Dissertation on "Spectral Methods for Shape Analysis"
Advisor: G. Orlandi. Co-advisor: U. Castellani

B.S. in Applied Mathematics — Sep. 2007 – Oct. 2010
**University of Verona** — *Verona, Italy*
Dissertation on "Existence and multiplicity of the solutions of the Plateau problem"
Advisor: S. Baldo

## Awards

- "Best overall method" for 6D detection of unseen objects — BOP Challenge 2024
- "Best overall method" for 6D localization of unseen objects — BOP Challenge 2024
- "Early-bird winner" of 6D detection of unseen objects — BOP Challenge 2024
- "Early-bird winner" of 6D localization of unseen objects — BOP Challenge 2024
- "Best method on TUD-L dataset" for the 6D localization of unseen objects — BOP Challenge 2023

## Publications

**Foundational feature fusion for conditional flow matching in 6D pose estimation** — BMVC 2026
A. Hamza, D. Boscaini, F. Poiesi
*Flow matching can estimate 6D poses of seen objects from frozen foundational features*

**Generative 6D pose estimation via conditional flow matching** — ICIP 2026
A. Hamza, D. Boscaini, W. Li, B. Busam, F. Poiesi
*First application of conditional flow matching to 6D pose estimation of seen objects*

**Action-guided generation of 3D functionality segmentation data** — CVPR-W 2026
J. Corsetti, F. Giuliari, D. Boscaini, P. Hermosilla, A. Pilzer, G. Mei, A. Delitzas, F. Engelmann, F. Poiesi
*Real-world 3D functionality segmentation data are scarse? Not a big deal, we can generate them!*

**High-resolution open-vocabulary object 6D pose estimation** — TPAMI, 2026
J. Corsetti, D. Boscaini, F. Giuliari, C. Oh, A. Cavallaro, F. Poiesi

**CHIP: A multi-sensor dataset for 6D pose estimation of chairs in industrial settings** — BMVC 2025
M. Nardon, M. Mujika, A. González, D. Sedano, J. Rueda, A. Caro, A. Caraffa, F. Poiesi, P.I. Chippendale, D. Boscaini

**Distilling 3D distinctive local descriptors for 6D pose estimation** IROS 2025
A. Hamza, A. Caraffa, D. Boscaini, F. Poiesi

**AI-driven visual monitoring of industrial assembly tasks** ICIAP 2025
M. Nardon, S. Messelodi, A. Granata, F. Poiesi, A. Danese, D. Boscaini

**Functionality understanding and segmentation in 3D scenes** CVPR 2025
J. Corsetti, F. Giuliari, A. Fasoli, D. Boscaini, F. Poiesi
*Highlight poster (top 3%). Awarded "Second place" for "Task-driven affordance grounding" at the SceneFun3D Challenge 2025*

**3D part segmentation via geometric aggregation of 2D visual features** WACV 2025
M. Garosi, R. Tedoldi, D. Boscaini, M. Mancini, N. Sebe, F. Poiesi

**Enabling seamless HRC integration: The AI-PRISM reference architecture** I-ESA 2024
M.A. Mateo-Casalí, J. Rueda, S.O. Afolaranmi, J.L.M. Lastra, D. Boscaini, L. Moya-Ruiz, F. Fraile

**Towards more effective human-robot collaboration via accurate pose estimation** ERF 2024
M.A. Mateo-Casalí, L. Moya-Ruiz, A. Caraffa, D. Boscaini, A. Hamza, P. Chippendale, F. Poiesi, F. Fraile

**Wild Berry image dataset collected in Finnish forests and peatlands using drones** ECCV-W 2024
L. Riz, S. Povoli, A. Caraffa, D. Boscaini, M.L. Mekhalfi, P. Chippendale, M. Turtiainen, B. Partanen, L.S. Ballester, F.B. Noguera, A. Franchi, E. Castelli, G. Piccinini, L. Marchesotti, M.S. Couceiro, F. Poiesi

**FreeZe: Training-free zero-shot 6D pose estimation with geometric and vision foundation models** ECCV 2024
A. Caraffa, D. Boscaini, A. Hamza, F. Poiesi
*An enhanced version of this work, FreeZe-v2, is the "early bird winner" in both 6D localization and 6D detection at the BOP Challenge 2024. An early version of this work, PoZe, won the "Best method on TUD-L dataset" award at the BOP Challenge 2023*

**Exploring fine-grained retail product discrimination with zero-shot object classification using Vision-Language Models** RTSI 2024
A. Tur, A. Conti, C. Beyan, D. Boscaini, R. Larcher, S. Messelodi, F. Poiesi, E. Ricci

**Open-vocabulary object 6D pose estimation** CVPR 2024
J. Corsetti, D. Boscaini, C. Oh, A. Cavallaro, F. Poiesi

**Tracciamento 3D della palla da punti di vista multipli nella pallavolo** Ital-IA 2024
L. Riz, S. Povoli, D. Boscaini, S. Messelodi, F. Poiesi

**Detect, Augment, Compose, and Adapt: Four steps for unsupervised domain adaptation in object detection** BMVC 2023
M.L. Mekhalfi, D. Boscaini, F. Poiesi

**Revisiting Fully Convolutional Geometric Features for object 6D pose estimation** ICCV-W 2023
J. Corsetti, D. Boscaini, F. Poiesi

**PatchMixer: Rethinking network design to boost generalization for 3D point cloud understanding** IMAVIS, 2023
D. Boscaini, F. Poiesi
*Novel network design that is intrinsically effective in generalisation across datasets unseen at training time*

**Supervised tractogram filtering using geometric deep learning** MIA, 2023
P. Astolfi, R. Verhagen, L. Petit, E. Olivetti, S. Sarubbo, J. Masci, D. Boscaini, P. Avesani

**The MONET dataset: Multimodal drone thermal dataset recorded in rural scenarios** CVPR-W 2023
L. Riz, A. Caraffa, M. Bortolon, M.L. Mekhalfi, D. Boscaini, A. Moura, J. Antunes, A. Dias, H. Silva, A. Leonidou, C. Constantinides, C. Keleshis, D. Abate, F. Poiesi

**Learning general and distinctive 3D local deep descriptors for point cloud registration** TPAMI, 2023
F. Poiesi, D. Boscaini
*State-of-the-art performance for point cloud registration in the transfer learning setting across 3DMatch, ETH, and Kitti datasets*

**SHIELD: Safeguard heritage in endangered looted districts** Ital-IA 2022
M.L. Mekhalfi, N. Saljoughi, D. Boscaini, F. Poiesi

**Localisation of defects in volumetric CT scans of valuable wood logs** ICPR-W 2020
D. Boscaini, F. Poiesi, S. Messelodi, A. Younes, D. Grande
*Selected for oral presentation*

**Joint supervised and self-supervised learning for 3D real-world challenges** ICPR 2020
A. Alliegro, D. Boscaini, T. Tommasi
*Selected for oral presentation (4.4% accaptance rate)*

**Distinctive 3D local deep descriptors** ICPR 2020
F. Poiesi, D. Boscaini

**Shape consistent 2D keypoint estimation under domain shift** ICPR 2020
L.O. Vasconcelos, M. Mancini, D. Boscaini, S. Rota Bulò, B. Caputo, E. Ricci

**Novel-view human action synthesis** ACCV 2020
M. Lakhal, D. Boscaini, F. Poiesi, O. Lanz, A. Cavallaro

**Clustered dynamic graph CNN for biometric 3D hand shape recognition** IJCB 2020
J. Svoboda, P. Astolfi, D. Boscaini, J. Masci, M.M. Bronstein

**Tractogram filtering of anatomically non-plausible fibers with geometric deep learning** MICCAI 2020
P. Astolfi, R. Verhagen, L. Petit, E. Olivetti, J. Masci, D. Boscaini, P. Avesani

**Self-supervision for 3D real-world challenges** ECCV-W 2020
A. Alliegro, D. Boscaini, T. Tommasi

**Deciphering interaction fingerprints from protein molecular surfaces** Nature Methods, 2020
P. Gainza, F. Sverrisson, F. Monti, E. Rodolà, D. Boscaini, M.M. Bronstein, B.E. Correira
*Advertised on the cover of the Feb 2020 issue of the journal*

**Learning interaction patterns from surface representations of protein structure** NeurIPS-W 2019
P. Gainza, F. Sverrisson, F. Monti, E. Rodolà, D. Boscaini, M.M. Bronstein, B.E. Correira

**Structured domain adaptation for 3D keypoint estimation** 3DV 2019
L.O. Vasconcelos, M. Mancini, D. Boscaini, B. Caputo, E. Ricci
*Oral presentation*

**3D shape segmentation with geometric deep learning** ICIAP 2019
D. Boscaini, F. Poiesi
*Spotlight presentation*

**Geometric deep learning on graphs and manifolds using mixture model CNNs** CVPR 2017
F. Monti*, D. Boscaini*, J. Masci, E. Rodolà, J. Svoboda, M.M. Bronstein
*Oral presentation (top 0.8%). First unified framework generalizing CNN architectures to non-Euclidean domains such as 3D shapes and graphs. Also available as technical report: arXiv:1611.08402. (* indicates equal contribution)*

**Geometric deep learning** SIGGRAPH Asia Courses 2016
J. Masci, E. Rodolà, D. Boscaini, M.M. Bronstein, H. Li

**Learning shape correspondence with anisotropic convolutional neural networks** NeurIPS 2016
D. Boscaini, J. Masci, E. Rodolà, M.M. Bronstein
*Presented also as a poster at the 3D Deep Learning Workshop (3DLL) 2016. Also available as technical report: arXiv:1605.06437*

**Anisotropic diffusion descriptors** CGF, 2016
D. Boscaini, J. Masci, E. Rodolà, M.M. Bronstein, D. Cremers
*Oral presentation at EUROGRAPHICS 2016*

**Geodesic convolutional neural networks on Riemannian manifolds** ICCV-W 2015
J. Masci*, D. Boscaini*, M.M. Bronstein, P. Vandergheynst
*Oral presentation at 3DRR 2015. First extension of the popular CNN paradigm to non-Euclidean domains. An early version of this work was published as the technical report arXiv:1501.06297 on January 2015. (* indicates equal contribution)*

**Learning class-specific descriptors for deformable shapes using localized spectral convolutional networks** CGF, 2015
D. Boscaini, J. Masci, S. Melzi, M.M. Bronstein, U. Castellani, P. Vandergheynst
*Oral presentation at SGP 2015*

**Shape-from-operator: Recovering shapes from intrinsic operators** CGF, 2015
D. Boscaini, D. Eynard, D. Kourounis, M.M. Bronstein
*Oral presentation at EUROGRAPHICS 2015. First approach able to synthesize the extrinsic geometry of a shape from intrinsic information. An early version of this work was published as the technical report arXiv:1406.1925 on June 2014*

**Coulomb shapes: Using electrostatic forces for deformation-invariant shape representation** EUROGRAPHICS-W 2014
D. Boscaini, R. Girdziusas, M.M. Bronstein
*Oral presentation at 3DOR 2014. Presented also as a poster at the International Computer Vision Summer School (ICVSS), 2014*

**A sparse coding approach for local-to-global 3D shape description** The Visual Computer, 2014
D. Boscaini, U. Castellani
*Invited paper. Journal extension of the 3DOR 2013 conference paper*

**Local signatures quantization by sparse coding** EUROGRAPHICS-W 2013
D. Boscaini, U. Castellani
*Oral presentation at 3DOR 2013. Presented also as a poster at SGP 2013*

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## Patents

**US patent application No. 17675011**
Clustered dynamic graph convolutional neural network for biometric 3D hand recognition
*Inventors: J. Svoboda, P. Astolfi, D. Boscaini, J. Masci*

**US patent No. 10210430** Filed Feb. 19, 2019
System and a method for learning features on geometric domains (CIP)
*Inventors: M.M. Bronstein, D. Boscaini, F. Monti • Acquired by Twitter Inc.*

**US patent No. 10013653** Filed Jul. 3, 2018
System and a method for learning features on geometric domains
*Inventors: M.M. Bronstein, D. Boscaini, J. Masci, P. Vandergheynst • Acquired by Twitter Inc.*

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## Invited talks

Object 6D pose estimation with vision and language Dec. 18, 2025
University of Catania, Italy • Invited by Francesco Ragusa, Giovanni Farinella

3D object understanding on the shoulders of 2D foundation models Jul. 29, 2025
Shandong University, China • Invited by Fabio Poiesi

Object 6D pose estimation in the foundation models era Nov. 22, 2024
University of Trento, Italy • Master course in "Trends and Applications in Computer Vision", Invited by Massimiliano Mancini

Object 6D pose estimation in the foundation models era Jun. 6, 2024
Politecnico di Torino, Italy • Invited by Francesca Pistilli

3D object understanding on the shoulders of 2D foundation models Mar. 28, 2024
École Polytechnique, Paris, France • Invited by Maks Ovsjanikov

3D deep learning to the test of real-world challenges Dec. 11, 2020
University of Pisa, Italy • Ph.D. event "Visions of Tomorrow"

3D Deep Learning Dec. 11, 2019
Politecnico di Torino, Italy • Invited by Tatiana Tommasi

Geometric deep learning for 3D shape analysis May 13, 2019
Politecnico di Torino, Italy • Invited by Barbara Caputo

Geometric deep learning for shape analysis Sep. 2, 2017
Kos, Greece • Keynote at EUSIPCO 2017

Geometric deep learning for shape analysis Apr. 4, 2017
FBK-TeV, Trento, Italy • Invited by Samuel Rota Bulò and Stefano Messelodi

Geometric deep learning for shape analysis — Feb. 13, 2017
CNR-IMATI, Genoa, Italy • Invited by Michela Spagnuolo

Deep learning on geometric data — Feb. 16, 2016
Zurich, Switzerland • SSSTC RiC big data research workshop

Deep learning on geometric data — Feb. 8, 2016
Embedded Vision Systems (eVS), Verona, Italy • Invited by Roberto Marzotto

Deep learning on geometric data — Feb. 4, 2016
Rainbow group, University of Cambridge, UK • Invited by Flora Tasse

Deep learning on geometric data — Feb. 3, 2016
University of Cambridge, UK • C.A.K.E. seminar, Invited by Simone Parisotto

Convolutional neural networks on non-Euclidean domains — Sep. 14, 2015
Potsdam, Germany • Keynote at SciCADE 2015

Shape-from-operators: recovering shapes from intrinsic differential operators — Nov. 26, 2014
TUM, Munich, Germany • Invited by Emanuele Rodolà

Shape-from-operators: recovering shapes from intrinsic differential operators — Aug. 19, 2014
Disentis, Switzerland • USI-ICS retreat

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## Teaching experience

### Academic courses

Trends and Applications in Computer Vision — University of Trento, Fall 2023

### Short courses and tutorials

Functional Maps: A Flexible Representation for Learning and Computing Correspondences — 3DV 2018
Geometric Deep Learning — SIGGRAPH Asia 2016
Deep Learning for Shape Analysis — EUROGRAPHICS 2016

### Teaching Assistantships

Computer Vision and Pattern Recognition — Università della Svizzera italiana, Spring 2017
Computer Vision and Pattern Recognition — Università della Svizzera italiana, Spring 2016
Large Scale Optimization — Università della Svizzera italiana, Spring 2016
Computer Graphics — Università della Svizzera italiana, Fall 2014
Geometric Image Processing and Computer Vision — Università della Svizzera italiana, Spring 2014
Calculus — Università della Svizzera italiana, Fall 2013
Mathematical Analysis 1, Mathematical Analysis 2 — University of Verona, 2012–2013
Mathematical Analysis 1, Mathematical Analysis 2 — University of Verona, 2011–2012
Mathematical Analysis 1 — University of Verona, 2010–2011

### Student supervision

Jaime Corsetti, PhD student at FBK and University of Trento — Nov. 2023–present
Project: Vision-language models for embodied AI

Mattia Nardon, Master student at University of Trento — Mar.–Dec. 2024
Role: Internship and Master thesis advisor
Project: AI-powered visual monitoring of Lego assembly tasks

Alice Fasoli, Master student at University of Trento — Mar.–Dec. 2024
Role: Internship and Master thesis advisor
Project: Retrieval-driven 6D pose estimation of unseen objects

Matteo Minardi, Master student at University of Trento — Mar.–Oct. 2024
Role: Internship and Master thesis advisor
Projects: Eye-gaze estimation using smart glasses, Vision encoder role in VLMs

Marco Garosi, Master student at University of Trento — Dec. 2023–Apr. 2024
Project: Zero-shot semantic segmentation of 3D objects
Outcome: Publication at WACV 2025

Riccardo Tedoldi, Master student at University of Trento — Dec. 2023–Apr. 2024

Project: Zero-shot semantic segmentation of 3D objects
Outcome: Publication at WACV 2025

Jaime Corsetti, Master student at University of Trento — 2022–Oct. 2023
Projects: Open-vocabulary and Supervised object 6D pose estimation for RGBD images
Outcome: Publication at CVPR 2024

Safa Abbes, Master student at University of Trento — 2022–2023
Role: Masther thesis coadvisor
Project: Self-supervised domain adaptation for RGB images

Antonio Alliegro, PhD student at Politecnico di Torino — 2020–2021
Project: Self-Supervised domain adaptation for 3D point clouds
Outcome: Publications at ECCV-W 2020 and ICPR 2021

Pietro Astolfi, PhD student at FBK, UniTN, and IIT — 2019–2021
Project: Geometric Deep Learning for brain structure analysis
Outcome: Publications at MICCAI 2020, IJCB 2020, and MIA 2023

Levi O. Vasconcelos, PhD student at UniTN and IIT — 2019–2020
Project: Structured domain adaptation
Outcome: Publications at 3DV 2019 and ICPR 2020

Antonio Alliegro, Master student at Politecnico di Torino — 2019–2020
Role: Masther thesis coadvisor

Piero Cavalcanti, Master student at Politecnico di Torino — 2019–2020
Role: Masther thesis coadvisor

Myriam Bronstein, Master student at Università della Svizzera italiana — 2016
Project: Machine learning methods on manifolds and graphs

Fatemeh Chegini, Master student at Università della Svizzera italiana — 2014–2015
Project: Spectral methods for cross-modal retrieval

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## Academic service

### Conferences revision activity

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|---|---|
| International Conference on Robotics and Automation (ICRA) | 2022, 2020 |
| International Conference on Pattern Recognition (ICPR) | 2022, 2020 |
| Symmetry and Geometry in Neural Representations (NeurIPS Workshops) | 2022 |
| International Conference on Image Analysis and Processing (ICIAP) | 2022 |
| International Conference on Machine Learning, Optimization, and Data Science (LOD) | 2022 |
| Symposium On Applied Computing (SAC) | 2022 |
| International Conference on 3D Vision (3DV) | 2021, 2020, 2019, 2018 |
| International Conference on Machine Learning, Optimization, and Data Science (LOD) | 2021 |
| International Conference on Machine Vision Applications (MVA) | 2021, 2019 |
| EUROGRAPHICS | 2019, 2017, 2015 |
| The British Machine Vision Conference (BMVC) | 2018 |
| Computer Vision and Pattern Recognition (CVPR) | 2017 |
| International Symposium on Vision, Modeling and Visualization (VMV) | 2016 |
| Neural Information Processing Systems (NeurIPS) | 2016 |

### Journal revision activity

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|---|---|
| Robotics and Automation Letters (RAL) | 2022 |
| Computer Graphics Forum (CGF) | 2022 |
| IEEE Transactions on Image Processing (TIP) | 2022, 2021 |
| IEEE Transactions on Transactions on Knowledge and Data Engineering (TKDE) | 2022, 2021 |
| Neural Processing Letters (NEPL) | 2022 |
| IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI) | 2021, 2020 |
| IEEE Transactions on Visualization and Computer Graphics (TVCG) | 2020, 2018, 2017 |
| Computers and Graphics | 2019 |

Computer Vision and Image Understanding (CVIU) 2019, 2015
International Journal of Machine Learning and Cybernetics (JMCL) 2019
Pattern Recognition Letters 2019
The Visual Computer Journal (TVCJ) 2018, 2017, 2016
Computer Aided Geometric Design (CAGD) 2018
Computer-Aided Design (CAD) 2018
Sensors 2018
IPSJ Transactions on Computer Vision and Applications 2017

### Area chair

British Machine Vision Conference (BMVC) 2025
British Machine Vision Conference (BMVC) 2024

### Program committee

International Workshop on Advances in Drone Vision (ADV) 2025
Graph Models for Learning and Recognition (GMLR) 2022
Organized within the 37th ACM Symposium on Applied Computing, Brno (Czech Republic)